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18/123,183	03/17/2023	Yu Fang	SONA-0010-U01-C01-C05	2646

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EXAMINER	
ALGIBHAH, HAMZA N	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/123,183	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2457	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/26/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,8-19 and 23-32 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,8-19 and 23-32 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1, 8-19, and 23-32*** are pending.

2. ***Claims 1, 8-19, and 23-32 are rejected.***

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. ***Claims 1, 8-11, 16, 19, 23-25 and 31-32*** *are rejected* under 35 U.S.C. 103) as being unpatentable over UNO et al (Pub. No.: US 2017/0330391 A1) in view of Miners et al (Pub. No.: US 2017/0026196 A1).

As per claim 1, UNO discloses an apparatus comprising:

- ***a policy acquisition circuit structured to interpret a vehicle policy data value comprising at least a portion of a vehicle policy for a vehicle*** (UNO, Fig 1 item 310, paragraph 0035-0036, 0056-0057, wherein "The present embodiment includes subject vehicles, a center that manages traveling information of the vehicles, and an external terminal, which is operated, for example, by an engineer to instruct collection of vehicle data from the vehicles via the center. The external terminal is connected to the center, for example, via an Internet connection and allows a collection condition for vehicle data from the vehicles to be input to the center. The collection condition for vehicle data is preferably set with a high flexibility. In the present embodiment, such highly flexible setting of the collection condition is achieved through inputting of scripts. Scripts refer to strings that describe one or more commands, or instructions, to be executed when vehicle data is collected. When a script input through the external terminal

is delivered to a vehicle via the center through wireless communication, commands described in the delivered script are executed on the part of the vehicle, so that vehicle data is collected based on the condition described in the script"; thus, the components within 'The in-vehicle communication device 310', inside the vehicle 300, that are responsible for receiving, storing, analyzing and interpreting the script (a vehicle policy data value) can be the policy acquisition circuit structured as claimed)***;***

- ***a policy processing circuit structured to generate, in response to and based at least in part on the vehicle policy data value, parsed policy data that comprises of one or more vehicle sub-policies of the vehicle policy*** (UNO, Fig 3, paragraph 0035-0042, 0057 wherein "Scripts refer to strings that describe one or more commands, or instructions, to be executed when vehicle data is collected ... the present embodiment, a script is capable of describing the following items (a) to (e). (a) Conditional expressions, which include a conditional expression for assessing the vehicle inside environment and a conditional expression that is related to expected situations and is used for identifying the vehicle outside environment; (b) A checking cycle, at which it is determined whether the vehicle data satisfies a conditional expression and images are classified according to the type of scene and checked; (c) A data collecting period from the start of data collection to the end; (d) A sampling cycle, at which data is collected into the storage; and (e) Vehicle data, which includes CAN data to be collected and values inside ECUs"; Thus, the parsed policy data can be conditional expressions, checking cycle, data collecting period, and/or sampling cycle)***; and***

- *determine from the vehicle policy data value ~~a type value of the vehicle policy~~, and a data resolution value* (UNO, Fig 3, paragraph 0040, 0046 wherein "A sampling cycle, at which data is collected into the storage"),

and

- ***the vehicle data resolution value defines, in part, a resolution at which at least one of ~~the passive~~ vehicle data or ~~the active~~ vehicle data is to be collected*** (UNO, paragraph 0040, 0052-0054, wherein "A sampling cycle, at which data is collected into the storage")***; and***

- ***a policy execution circuit structured to collect vehicle data from one or more vehicle sensors in response to the parsed policy data*** (UNO, paragraph 0057, 0080-0085, wherein "The vehicle 300 includes an in-vehicle communication device 310, which wirelessly transmits and receives various kinds of information including the outbound message to and from the center 100. When receiving a message sent from the center 100 via a communication section 311, the in-vehicle communication device 310 extracts a script from the received message and temporarily stores the script in a script storage section 312. The in-vehicle communication device 310 then inputs the stored script to a vehicle-side script analyzing section 313. The communication section 311 of the in-vehicle communication device 310, the script storage section 312, the vehicle-side script analyzing section 313, a script interpretation section 314, a vehicle network communication section 315, and a vehicle data storage section 316 can be constituted by various types of circuits. For example, these sections may be configured by a communication circuit, a script storage circuit, a vehicle-side script analyzing circuit, a script interpretation circuit, a vehicle network communication circuit, and a vehicle data storage circuit ... The script interpretation section 314 sequentially reads and executes the instructions in the script, thereby collecting vehicle data based on the condition designated by a script via the vehicle network communication section 315").

UNO does not explicitly disclose *a type value of the vehicle policy; wherein the type value is at least one of a passive vehicle policy, wherein the vehicle policy specifies passive vehicle data to be initially collected from one or more vehicle sensors during ordinary operations of the vehicle; or an active vehicle policy, wherein the vehicle policy specifies active vehicle data to*

be initially collected from the ore more vehicle sensors responsive to a request for the vehicle data transmitted to one or more endpoints, wherein the request is based at least in part on the vehicle policy;. However, Miners discloses

a type value of the vehicle policy; wherein the type value is at least one of a passive vehicle policy (Miners, paragraph 0002-0003, 0011 wherein "An on-vehicle device for a driver monitoring system can be configured to be isolated from the vehicle bus while connected to the vehicle's OBD port. In a fully-isolated mode, the device only receives power and ground from the vehicle's OBD port and there is no other communication sent or received from the vehicle bus. In a passive mode, the device can obtain some information from the vehicle bus but does not request any information on the vehicle bus ... Optionally, the device receives commands via a communication interface to be switched among the three modes: fully active mode, fully isolated mode and passive mode"),

- *wherein the vehicle policy specifies passive vehicle data to be initially collected from one or more vehicle sensors during ordinary operations of the vehicle* (Miners, paragraph 0005, 0007, 0014, wherein "In passive-only mode, the device 12 may also receive power and ground from the OBD port 40. In passive-only mode, the device 12 receives data passively from the OBD port 40 and vehicle bus 42, such as by "sniffing" data on the bus 42. In this way, the device 12 receives from the vehicle bus 42 data indicative of vehicle engine operating parameters such as vehicle speed, engine speed, temperature, fuel consumption (or electricity consumption), engine idle time, car diagnostics (from OBD) and other information that is related to mechanical operation of the vehicle")*; or*

- *an active vehicle policy, wherein the vehicle policy specifies active vehicle data to be initially collected from the ore more vehicle sensors responsive to a request for the vehicle data transmitted to one or more endpoints* (Miners, paragraph 0003, wherein "Optionally, the

device receives commands via a communication interface to be switched among the three modes: fully active mode, fully isolated mode and passive mode. The commands may come from a server and may be communicated over the internet or over a cell network to a communication module on the device").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to incorporate Miners selectable passive and active modes into UNO's collection script such that the collection script indicates passive or active mode for collecting the data which will result in making the request for the data based at least in part on the vehicle policy as claimed because this would have provided a way to reduce and/or eliminate unnecessary active communication when data can be collected passively and thus improving the performance of the system.

As pre claim 8, claim 1 is incorporated and UNO in view of Miners discloses **wherein the policy execution circuit is further structured to transmit a begin collection command value to actively collect the vehicle data** (UNO, paragraph 0035, 0066, wherein "Scripts refer to strings that describe one or more commands, or instructions, to be executed when vehicle data is collected ... As a result, the vehicle-side script analyzing section 313 delivers, as a normal reply, the analyzed script to the script interpretation section 314 to start collection of the vehicle data based on the condition designated by the script"; Miners, paragraph 0003, wherein "Optionally, the device receives commands via a communication interface to be switched among the three modes: fully active mode, fully isolated mode and passive mode").

As pre claim 9, claim 1 is incorporated and UNO in view of Miners discloses **wherein the policy execution circuit is further structured to generate, based at least in part on the**

collected vehicle data, a vehicle property value to actively collect the vehicle data (UNO, paragraph 0035, 0066, wherein "Scripts refer to strings that describe one or more commands, or instructions, to be executed when vehicle data is collected ... As a result, the vehicle-side script analyzing section 313 delivers, as a normal reply, the analyzed script to the script interpretation section 314 to start collection of the vehicle data based on the condition designated by the script"; Miners, paragraph 0003, wherein "Optionally, the device receives commands via a communication interface to be switched among the three modes: fully active mode, fully isolated mode and passive mode").

As pre claim 10, claim 1 is incorporated and UNO in view of Miners discloses wherein the policy execution circuit is further structured to transmit a query value to actively collect the vehicle data (UNO, paragraph 0035, 0066, wherein "Scripts refer to strings that describe one or more commands, or instructions, to be executed when vehicle data is collected ... As a result, the vehicle-side script analyzing section 313 delivers, as a normal reply, the analyzed script to the script interpretation section 314 to start collection of the vehicle data based on the condition designated by the script"; Miners, paragraph 0003, claim 2, wherein "Optionally, the device receives commands via a communication interface to be switched among the three modes: fully active mode, fully isolated mode and passive mode").

As pre claim 11, claim 1 is incorporated and UNO discloses a memory device structured to store the collected vehicle data (UNO, Fig 3, paragraph 0085, wherein "As shown in FIG. 1, the vehicle network communication section 315 stores the vehicle data collected via the vehicle network NW in a vehicle data storage section 316, which functions as the above described storage").

As pre claim 16, claim 1 is incorporated and UNO discloses **a collected data provisioning circuit structured to transmit the collected vehicle data** (UNO, Fig 3, paragraph 0026-0027, wherein the filtered data can then be stored locally (516) or uploaded to an external device, server or repository (518)).

Claims *19, 23-25 and 31-32* are rejected under the same rationale as claim *1, 8-11, and 16*.

4. ***Claims 12-13. 17-18 and 26-27*** *are rejected* under 35 U.S.C. 103) as being unpatentable over UNO et al (Pub. No.: US 2017/0330391 A1) in view of Miners et al (Pub. No.: US 2017/0026196 A1) and BARRETT et al (Pub. No.: US 2019/0394089 A1).

As per claim 12, claim 1 is incorporated and UNO and Miners do not explicitly disclose ***a system, comprising a converged network device (CND) structured to regulate communications between a first network zone and a second network zone, the first network zone having a first vehicle sensor of the one or more vehicle sensors and the second network zone having a second vehicle sensor of the one or more vehicle sensors.*** However, BARRETT discloses ***a system, comprising a converged network device (CND) structured to regulate communications between a first network zone and a second network zone, the first network zone having a first vehicle sensor of the one or more vehicle sensors and the second network zone having a second vehicle sensor of the one or more vehicle sensors*** (BARRETT, Fig 1, 2, paragraph 0021, 0027-0028 vehicle network, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay. BARRETT, Fig 2,

paragraph 0037, wherein for example the gateway 240 can be the converged network device (CND) which is between internal network nodes of a first type (such as CAN) and telecoms control unit 260 (the transceiver) and network nodes of a second type (such as LIN and/or FlexRay)).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to incorporate BARRETT to UNO and Miners to achieve the claimed limitations because this would have provided a way to apply vehicle data collection policies to vehicles having sensors distributed in multiple network zones within the vehicle.

As pre claim 13, claim 12 is incorporated and BARRETT discloses **wherein the first network zone and the second network zone are of distinct types** (BARRETT, Fig 1, 2, paragraph 0021, 0027-0028 vehicle network, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay. BARRETT, Fig 2, paragraph 0037, wherein for example the gateway 240 can be the converged network device (CND) which is between internal network nodes of a first type (such as CAN) and telecoms control unit 260 (the transceiver) and network nodes of a second type (such as LIN and/or FlexRay));

As per claim 17, claim 1 is incorporated and UNO and Miners do not explicitly disclose ***wherein the vehicle policy data value is encrypted.*** BARRETT already discloses encrypting the collected/filtered data (UNO, Fig 3, paragraph 0020, wherein the filtering or redaction of PII may be performed on individual objects within an image or video stream such as for example the license plate 110 to generate an image 142 using an encryption key 130 to generate a redacted

111 masked object 141. The object may be masked within the original image 142, or extracted to a separate file 143 containing the masked object 141 which can be encrypted and stored separately. The encryption process may utilize a two-step encryption process to encrypt the content) and encrypting data is well known in the art.

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify UNO so that the vehicle policy data value is encrypted as claimed because this would have provided a way to improve the security of the system by protecting the policy.

As per claim 18, claim 1 is incorporated and UNO and Miners do not explicitly disclose **wherein the parsed policy data is encrypted.** BARRETT already discloses encrypting the collected/filtered data (UNO, Fig 3, paragraph 0020, wherein the filtering or redaction of PII may be performed on individual objects within an image or video stream such as for example the license plate 110 to generate an image 142 using an encryption key 130 to generate a redacted 111 masked object 141. The object may be masked within the original image 142, or extracted to a separate file 143 containing the masked object 141 which can be encrypted and stored separately. The encryption process may utilize a two-step encryption process to encrypt the content) and encrypting data is well known in the art.

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify UNO so that the parsed policy data is encrypted as claimed because this would have provided a way to improve the security of the system by protecting the policy.

Claims ***26-27*** are rejected under the same rationale as claims 12-13.

5. ***Claims 14-15 and 28-29*** *are rejected* under 35 U.S.C. 103) as being unpatentable over UNO et al (Pub. No.: US 2017/0330391 A1) in view of Miners et al (Pub. No.: US 2017/0026196 A1) and HASEGAWA (Pub. No.: US 2017/0352261 A1).

As per claim 14, claim 1 is incorporated and UNO and Miners do not explicitly disclose ***wherein the policy execution circuit is further structured to delegate collection of the vehicle data to one or more vehicle controllers via transmitting at least some of the parsed policy data to the one or more vehicle controllers.*** However, HASEGAWA discloses ***wherein the policy execution circuit is further structured to delegate collection of the vehicle data to one or more vehicle controllers via transmitting at least some of the parsed policy data to the one or more vehicle controllers*** (HASEGAWA, paragraph 0043, wherein "the in-vehicle communication device 310 requests a determination on the vehicle environment to vehicle ECUs that are the sources of the environment information described as conditional expressions in a script, which information is the vehicle inside environment information or the vehicle outside environment information. For example, if the script describes a lane change as a conditional expression, the in-vehicle communication device 310 requests the image processing ECU 342, which is capable of determining whether there has been a lane change, to start the determination. If a determination result indicating that the environment information described in the script agrees with the vehicle environment in which the vehicle 300 is situated is delivered to the in-vehicle communication device 310 from the vehicle ECU that has made the determination, the in-vehicle communication device 310 requests the vehicle ECU that is the source of the vehicle data described in the script to collect vehicle data.").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to incorporate HASEGAWA to UNO and Miners to achieve

the claimed limitations because this would have provided a way to improve the performance of the system by benefiting from using ECUs already configured to collect the required data.

As pre claim 15, claim 14 is incorporated and HASEGAWA discloses ***a collected data acquisition circuit structured to interpret the vehicle data collected by the one or more vehicle controllers*** (HASEGAWA, paragraph 0043, wherein "the in-vehicle communication device 310 requests a determination on the vehicle environment to vehicle ECUs that are the sources of the environment information described as conditional expressions in a script, which information is the vehicle inside environment information or the vehicle outside environment information. For example, if the script describes a lane change as a conditional expression, the in-vehicle communication device 310 requests the image processing ECU 342, which is capable of determining whether there has been a lane change, to start the determination. If a determination result indicating that the environment information described in the script agrees with the vehicle environment in which the vehicle 300 is situated is delivered to the in-vehicle communication device 310 from the vehicle ECU that has made the determination, the in-vehicle communication device 310 requests the vehicle ECU that is the source of the vehicle data described in the script to collect vehicle data.")

Claims ***28-29*** are rejected under the same rationale as claims 14-15.

Response to Arguments

Applicant's arguments filed 08/24/2026 have been fully considered but they are now moot in light of the new grounds of rejection.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ario Etienne can be reached at ario.etienne@uspto.gov. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2457